

# Лабораторная работа №7

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Получение навыков настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.

# Выполнение лабораторной работы

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## 1. Я запущу GNS3 VM и GNS3. Создам новый проект.

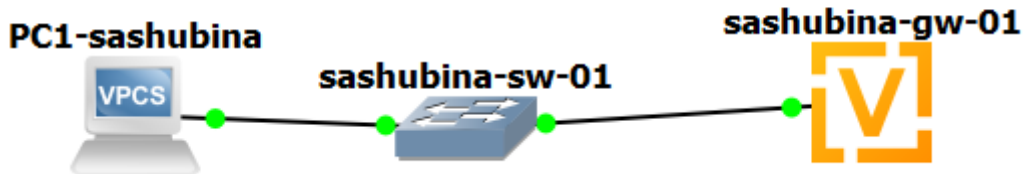


Рис. 1: новый проект

Я включу захват трафика на соединении между коммутатором sw-01 и маршрутизатором gw-01.

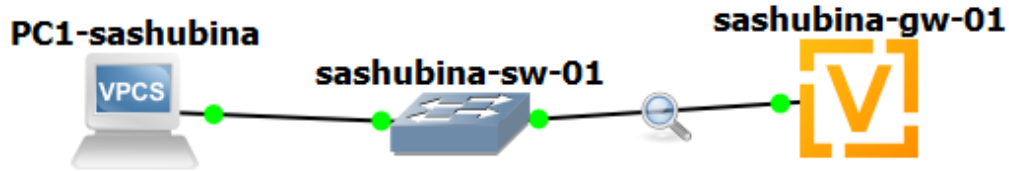


Рис. 2: захват трафика

# Настройка VyOS:

```
vyos login: vyos
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ install image
You are trying to install from an already installed system. An ISO
image file to install or URL must be specified.
Exiting...
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name sashubina-gw-01
[edit]
vyos@vyos# set system domain-name sashubina.net
[edit]
```

Рис. 3: установка имени

```
456s@vyos# set system login user sashubina authentication plaintext-password 123
[edit]
vyos@vyos# show system login user sashubina
+authentication {
+  plaintext-password 123456
+}
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ exit
logout

Welcome to VyOS - sashubina-gw-01 ttyS0

sashubina-gw-01 login: sashubina
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
sashubina@sashubina-gw-01:~$
```

Рис. 4: сохранение

После этого я выполню вход под созданным пользователем и удалю пользователя vyos:

```
sashubina@sashubina-gw-01:~$ configure
[edit]
sashubina@sashubina-gw-01# delete system login user vyos
[edit]
sashubina@sashubina-gw-01# commit
[edit]
sashubina@sashubina-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
sashubina@sashubina-gw-01#
```

Рис. 5: удаление



На маршрутизаторе, под созданным пользователем в режиме конфигурирования, я настрою адресацию IPv4:

```
sashubina@sashubina-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24  
[edit]
```

Рис. 6: настройка адресации

## Я добавлю конфигурацию DHCP-сервера на маршрутизаторе:

```
sashubina@sashubina-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24
[edit]
domain-name username.net^C  configure^C
[edit]
domain-name sashubina.net set service dhcp-server shared-network-name sashubina
[edit]
name-server 10.0.0.1w-01# set service dhcp-server shared-network-name sashubina
[edit]
subnet 10.0.0.0/24 default-router 10.0.0.1-server shared-network-name sashubina
[edit]
subnet 10.0.0.0/24 range hosts start 10.0.0.2rver shared-network-name sashubina
[edit]
subnet 10.0.0.0/24 range hosts stop 10.0.0.253ver shared-network-name sashubina
[edit]
```

Рис. 7: добавление конфигурации

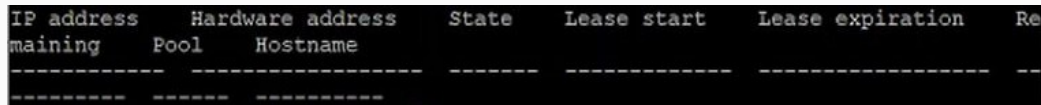
```
sashubina@sashubina-gw-01# commit
[edit]
sashubina@sashubina-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
sashubina@sashubina-gw-01# exit
exit
sashubina@sashubina-gw-01:~$
```

Рис. 8: добавление конфигурации

Для просмотра статистики DHCP-сервера и выданных адресов я буду использовать команды:

```
sashubina@sashubina-gw-01:~$ show dhcp server statistics
Pool          Size    Leases    Available  Usage
-----
sashubina      252         0         252    0%
sashubina@sashubina-gw-01:~$ show dhcp server leases
IP address    Hardware address    State    Lease start    Lease expiration    Remaining    Pool    Hostname
-----
```

Рис. 9: просмотр статистики



| IP address | Hardware address | State    | Lease start | Lease expiration | Re  |
|------------|------------------|----------|-------------|------------------|-----|
| maining    | Pool             | Hostname |             |                  |     |
| -----      | -----            | -----    | -----       | -----            | --- |
| -----      | -----            | -----    |             |                  |     |

Рис. 10: просмотр статистики

# Я настрою оконечное устройство PC1:

```
VPCS> dhcp -d
Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS1
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:
68:00

Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS1
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:
68:00

Opcode: 2 (REPLY)
Client IP Address: 0.0.0.0
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Offer
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = sashubina.net
```

Рис. 11: настройка оконечного устройства

# настройка оконечного устройства

```
Opcode: 1 (REQUEST)
Client IP Address: 10.0.0.2
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Request
Option 54: DHCP Server = 10.0.0.1
Option 50: Requested IP Address = 10.0.0.2
Option 61: Client Identifier = Hardware Type-Ethernet MAC Address = 00:50:79:66:
68:00
Option 12: Host Name = VPCS1

Opcode: 2 (REPLY)
Client IP Address: 10.0.0.2
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Ack
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = sashubina.net

IP 10.0.0.2/24 GW 10.0.0.1

VPCS> save
Saving startup configuration to startup.vpc
. done
VPCS> █
```

Рис. 12: настройка оконечного устройства

## Я проверю конфигурацию IPv4 на узле и пропингую маршрутизатор:

```
VPCS> show ip  
  
NAME           : VPCS[1]  
IP/MASK        : 10.0.0.2/24  
GATEWAY        : 10.0.0.1  
DNS            : 10.0.0.1  
DHCP SERVER    : 10.0.0.1  
DHCP LEASE     : 86167, 86400/43200/75600  
DOMAIN NAME    : sashubina.net  
MAC            : 00:50:79:66:68:00  
LPORT          : 10003  
RHOST:PORT     : 127.0.0.1:10004  
MTU:           : 1500
```

Рис. 13: проверка конфигурации



```
VPCS> ping 10.0.0.1 -c 2  
84 bytes from 10.0.0.1 icmp_seq=1 ttl=64 time=1.245 ms  
84 bytes from 10.0.0.1 icmp_seq=2 ttl=64 time=0.969 ms  
VPCS> █
```

Рис. 14: пинг

На маршрутизаторе я вновь посмотрю статистику DHCP-сервера и выданные адреса. В отчёте я поясню полученную информацию:

```
sashubina@sashubina-gw-01:~$ show dhcp server statistics
Pool      Size    Leases   Available  Usage
-----
sashubina 252      1        251      0%
sashubina@sashubina-gw-01:~$ show dhcp server leases
IP address  Hardware address  State  Lease start  Lease expiration  Remaining  Pool  Hostname
-----
10.0.0.2    00:50:79:66:68:00  active  2025/11/28 19:08:51  2025/11/29 19:08:51  23:53:57  sashubina  VPCS1
sashubina@sashubina-gw-01:~$
```

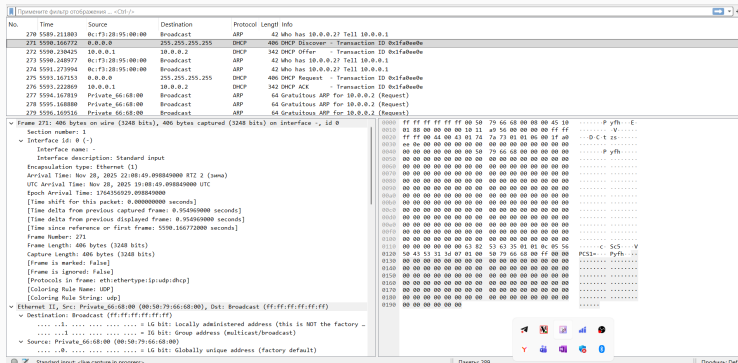
Рис. 15: просмотр статистики

# На маршрутизаторе я посмотрю журнал работы DHCP-сервера:

```
sshubina@sshubina-pe-01:~$ show log | grep dhcp
Nov 28 17:36:41 dhclient-script-vyos[2285]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:36:42 vyos-hostsd[615]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:36:42 dhclient-script-vyos[2285]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:36:42 vyos-hostsd[615]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:38:19 dhclient-script-vyos[2492]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:38:19 vyos-hostsd[615]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:38:19 dhclient-script-vyos[2492]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:38:19 vyos-hostsd[615]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:40:15 dhclient-script-vyos[2547]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:40:15 vyos-hostsd[615]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:40:15 dhclient-script-vyos[2547]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:40:15 vyos-hostsd[615]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:42:29 dhclient-script-vyos[2601]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:42:29 vyos-hostsd[615]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:42:29 dhclient-script-vyos[2601]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:42:29 vyos-hostsd[615]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:44:05 dhclient-script-vyos[2652]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:44:05 vyos-hostsd[615]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:44:05 dhclient-script-vyos[2652]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:44:05 vyos-hostsd[615]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:44:06 dhclient-script-vyos[2702]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:44:06 vyos-hostsd[615]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:45:53 dhclient-script-vyos[2737]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:45:53 vyos-hostsd[615]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:45:54 dhclient-script-vyos[2737]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:45:54 vyos-hostsd[615]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:47:53 dhclient-script-vyos[2774]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:47:53 vyos-hostsd[615]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:47:53 dhclient-script-vyos[2774]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:47:53 vyos-hostsd[615]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:49:26 dhclient-script-vyos[2774]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:49:26 vyos-hostsd[615]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:49:26 dhclient-script-vyos[2774]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:49:26 vyos-hostsd[615]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:50:58 dhclient-script-vyos[2800]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:50:58 vyos-hostsd[615]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:50:58 dhclient-script-vyos[2800]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:50:58 vyos-hostsd[615]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:52:35 dhclient-script-vyos[2843]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:52:35 vyos-hostsd[615]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:52:35 dhclient-script-vyos[2843]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:52:35 vyos-hostsd[615]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Nov 28 17:54:51 dhclient-script-vyos[2877]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Nov 28 17:54:51 vyos-hostsd[615]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
```

Рис. 16: просмотр журнала работы

# В отчёте я проанализирую захваченные анализатором трафика пакеты, относящиеся к работе DHCP и назначению адреса устройству.



The screenshot shows the Wireshark interface with a list of captured packets and a detailed view of a selected packet. The packet list on the left shows several DHCP-related packets, including a DHCP Discover (342) and a DHCP Offer (342). The packet details pane on the right shows the structure of the selected packet, including the Ethernet II header, Internet Protocol Version 4 header, and the DHCP message structure. The packet bytes pane on the far right shows the raw data in hexadecimal and ASCII.

| No. | Time        | Source            | Destination     | Protocol | Length | Info                                      |
|-----|-------------|-------------------|-----------------|----------|--------|-------------------------------------------|
| 270 | 5589.211803 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42     | Who has 10.0.0.2? Tell 10.0.0.1           |
| 271 | 5590.166772 | 0.0.0.0           | 255.255.255.255 | DHCP     | 406    | DHCP Discover - Transaction ID 0x1fab0e0e |
| 272 | 5590.230425 | 10.0.0.1          | 10.0.0.2        | DHCP     | 342    | DHCP Offer - Transaction ID 0x1fab0e0e    |
| 273 | 5590.248977 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42     | Who has 10.0.0.2? Tell 10.0.0.1           |
| 274 | 5591.273994 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42     | Who has 10.0.0.2? Tell 10.0.0.1           |
| 275 | 5593.167153 | 0.0.0.0           | 255.255.255.255 | DHCP     | 406    | DHCP Request - Transaction ID 0x1fab0e0e  |
| 276 | 5593.222869 | 10.0.0.1          | 10.0.0.2        | DHCP     | 342    | DHCP ACK - Transaction ID 0x1fab0e0e      |
| 277 | 5594.167819 | Private_66:68:00  | Broadcast       | ARP      | 64     | Gratuitous ARP for 10.0.0.2 (Request)     |
| 278 | 5595.168880 | Private_66:68:00  | Broadcast       | ARP      | 64     | Gratuitous ARP for 10.0.0.2 (Request)     |
| 279 | 5596.169516 | Private_66:68:00  | Broadcast       | ARP      | 64     | Gratuitous ARP for 10.0.0.2 (Request)     |

▼ Frame 271: 406 bytes on wire (3248 bits), 406 bytes captured (3248 bits) on interface -, id 0

Section number: 1

▼ Interface 0 (-)

Interface description: Standard input

Encapsulation type: Ethernet (1)

Arrival Time: Nov 28, 2025 22:08:49.098849000 RTZ (3ms)

UTC Arrival Time: Nov 28, 2025 19:08:49.098849000 UTC

Epoch Arrival Time: 1764350929.098849000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 0.954960000 seconds]

[Time delta from previous displayed frame: 0.954960000 seconds]

[Time since reference or first frame: 5590.166772000 seconds]

Frame Number: 271

Frame Length: 406 bytes (3248 bits)

Capture Length: 406 bytes (3248 bits)

[Frame is marked: false]

[Frame is ignored: false]

[Protocols in frame: ethertype:ip:udp:dhcp]

[Coloring Rule Name: UDP]

[Coloring Rule String: udp]

▼ Ethernet II, Src: Private\_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

▼ Destination: Broadcast (ff:ff:ff:ff:ff:ff)

....0.. .... = LG bit: Locally administered address (this is NOT the factory ...)

....0.. .... = IG bit: Group address (multicast/broadcast)

▼ Source: Private\_66:68:00 (00:50:79:66:68:00)

....0.. .... = LG bit: Globally unique address (factory default)

Рис. 17: анализ трафика

### Рис. 18: анализ трафика

The screenshot displays the Wireshark network traffic analysis interface. The top pane shows a list of captured packets. The middle pane displays the details of the selected packet (Frame 272), and the bottom pane shows the raw packet data in hexadecimal and ASCII.

**Packet List:**

| No. | Time        | Source            | Destination     | Protocol | Length | Info                                     |
|-----|-------------|-------------------|-----------------|----------|--------|------------------------------------------|
| 270 | 5599.218803 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42     | Who has 10.0.0.2? Tell 10.0.0.1          |
| 271 | 5599.266772 | 0.0.0.0           | 255.255.255.255 | DHCP     | 406    | DHCP Discover - Transaction ID 0a1fa8e0e |
| 272 | 5599.290425 | 10.0.0.1          | 10.0.0.2        | DHCP     | 342    | DHCP Offer - Transaction ID 0a1fa8e0e    |
| 273 | 5599.248977 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42     | Who has 10.0.0.2? Tell 10.0.0.1          |
| 274 | 5599.273994 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42     | Who has 10.0.0.2? Tell 10.0.0.1          |
| 275 | 5599.167153 | 0.0.0.0           | 255.255.255.255 | DHCP     | 406    | DHCP Request - Transaction ID 0a1fa8e0e  |
| 276 | 5599.222809 | 10.0.0.1          | 10.0.0.2        | DHCP     | 342    | DHCP ACK - Transaction ID 0a1fa8e0e      |
| 277 | 5594.167819 | Private_66:68:00  | Broadcast       | ARP      | 64     | Gratuitous ARP for 10.0.0.2 (Request)    |
| 278 | 5595.168880 | Private_66:68:00  | Broadcast       | ARP      | 64     | Gratuitous ARP for 10.0.0.2 (Request)    |
| 279 | 5595.169516 | Private_66:68:00  | Broadcast       | ARP      | 64     | Gratuitous ARP for 10.0.0.2 (Request)    |

**Packet Details (Frame 272):**

- Section number: 1
- Interface id: 0 (-)
- Interface name: -
- Interface description: Standard input
- Encapsulation type: Ethernet (1)
- Arrival Time: Nov 28, 2025 22:08:49.162502000 RTZ 2 (jawa)
- UTC Arrival Time: Nov 28, 2025 19:08:49.162502000 UTC
- Epoch Arrival Time: 1764356929.162502000
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 0.063653000 seconds]
- [Time delta from previous displayed frame: 0.063653000 seconds]
- [Time since reference or first frame: 5599.230425000 seconds]
- Frame Number: 272
- Frame Length: 342 bytes (2736 bits)
- Capture Length: 342 bytes (2736 bits)
- [Frame is marked: false]
- [Frame is ignored: false]
- [Protocols in frame: ethernetII.typeIIip dhcp]
- [Coloring Rule Name: UDP]
- [Coloring Rule String: udp]
- Ethernet II, Src: 0c:f3:28:95:00:00 (0c:f3:28:95:00:00), Dst: Private\_66:68:00 (00:50:79:66:68:00)
- Destination: Private\_66:68:00 (00:50:79:66:68:00)
- ...:0 ..... = LG bit: Globally unique address (factory default)
- ...:0 ..... = IG bit: Individual address (unicast)
- Source: 0c:f3:28:95:00:00 (0c:f3:28:95:00:00)
- ...:0 ..... = LG bit: Globally unique address (factory default)

**Raw Packet Data:**

```
0000 00 50 79 66 68 00 0c f3 28 95 00 00 00 45 10 -P yth... (....E-
0010 01 48 00 00 00 00 11 25 93 0a 00 00 01 0a 00 -H.....
0020 00 02 00 41 00 4d 01 34 24 52 02 01 00 00 1f 40 -...C D 4 $.....
0030 0e 0e 00 00 00 00 00 00 00 0a 00 00 02 00 00 -.....P yth.....
0040 00 00 00 00 00 00 50 79 66 68 00 00 00 00 00 -.....
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 -.....
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 -.....
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 -.....
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 -.....
0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 -.....
00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 -.....
00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 -.....
00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 -.....
00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 -.....
00e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 -.....
00f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 -.....
0100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 -.....
0110 00 00 00 00 00 63 82 53 63 35 01 02 16 64 0a -.....c Se5-G-
0120 00 00 01 23 04 00 01 51 00 01 04 ff ff ff 00 03 -...3-Q.....
0130 04 0a 00 00 01 05 04 0a 00 00 01 0f 0d 73 61 73 -hubina.n et.....
0140 68 75 62 69 6e 61 2e 6e 65 76 ff 00 00 00 00 00 -.....
0150 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 -.....
```

Рис. 19: анализ трафика

| Примените фильтр отображения: ... <Ctrl>                  |             |                   |                 |          |                                               |
|-----------------------------------------------------------|-------------|-------------------|-----------------|----------|-----------------------------------------------|
| No.                                                       | Time        | Source            | Destination     | Protocol | Length Info                                   |
| 270                                                       | 5589.211893 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42 Who has 10.0.0.2? Tell 10.0.0.1            |
| 271                                                       | 5590.166772 | 0.0.0.0           | 255.255.255.255 | DHCP     | 406 DHCP Discover - Transaction ID 0cfafab0e0 |
| 272                                                       | 5590.238425 | 10.0.0.1          | 10.0.0.2        | DHCP     | 342 DHCP Offer - Transaction ID 0cfafab0e0    |
| 273                                                       | 5590.248977 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42 Who has 10.0.0.2? Tell 10.0.0.1            |
| 274                                                       | 5591.273994 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42 Who has 10.0.0.2? Tell 10.0.0.1            |
| 275                                                       | 5593.167133 | 0.0.0.0           | 255.255.255.255 | DHCP     | 406 DHCP Request - Transaction ID 0cfafab0e0  |
| 276                                                       | 5593.222809 | 10.0.0.1          | 10.0.0.2        | DHCP     | 342 DHCP ACK - Transaction ID 0cfafab0e0      |
| 277                                                       | 5594.167819 | Private_66:68:00  | Broadcast       | ARP      | 64 Gratuitous ARP for 10.0.0.2 (Request)      |
| 278                                                       | 5595.168880 | Private_66:68:00  | Broadcast       | ARP      | 64 Gratuitous ARP for 10.0.0.2 (Request)      |
| 279                                                       | 5596.169516 | Private_66:68:00  | Broadcast       | ARP      | 64 Gratuitous ARP for 10.0.0.2 (Request)      |
| UDP payload (300 bytes)                                   |             |                   |                 |          |                                               |
| Dynamic Host Configuration Protocol (Offer)               |             |                   |                 |          |                                               |
| Message type: Boot Reply (2)                              |             |                   |                 |          |                                               |
| Hardware type: Ethernet (0x01)                            |             |                   |                 |          |                                               |
| Hardware address length: 6                                |             |                   |                 |          |                                               |
| Hops: 0                                                   |             |                   |                 |          |                                               |
| Transaction ID: 0cfafab0e0                                |             |                   |                 |          |                                               |
| Seconds elapsed: 0                                        |             |                   |                 |          |                                               |
| Bootp flags: 0x0000 (Unicast)                             |             |                   |                 |          |                                               |
| Client IP address: 0.0.0.0                                |             |                   |                 |          |                                               |
| Your (client) IP address: 10.0.0.2                        |             |                   |                 |          |                                               |
| Next server IP address: 0.0.0.0                           |             |                   |                 |          |                                               |
| Relay agent IP address: 0.0.0.0                           |             |                   |                 |          |                                               |
| Client MAC address: Private_66:68:00 (00:50:79:66:68:00)  |             |                   |                 |          |                                               |
| Client hardware address padding: 000000000000000000000000 |             |                   |                 |          |                                               |
| Server host name not given                                |             |                   |                 |          |                                               |
| Boot file name not given                                  |             |                   |                 |          |                                               |
| Magic cookie: DHCP                                        |             |                   |                 |          |                                               |
| Option: (53) DHCP Message Type (Offer)                    |             |                   |                 |          |                                               |
| Option: (54) DHCP Server Identifier (10.0.0.1)            |             |                   |                 |          |                                               |
| Option: (51) IP Address Lease Time                        |             |                   |                 |          |                                               |
| Option: (1) Subnet Mask (255.255.255.0)                   |             |                   |                 |          |                                               |
| Option: (3) Router                                        |             |                   |                 |          |                                               |
| Option: (6) Domain Name Server                            |             |                   |                 |          |                                               |
| Option: (15) Domain Name                                  |             |                   |                 |          |                                               |
| Option: (255) End                                         |             |                   |                 |          |                                               |
| Padding: 00000000000000000000000000000000                 |             |                   |                 |          |                                               |
| Standard input: <live capture in progress>                |             |                   |                 |          |                                               |
| Пакеты: 289                                               |             |                   |                 |          |                                               |
| Профиль: Default                                          |             |                   |                 |          |                                               |

Рис. 20: анализ трафика

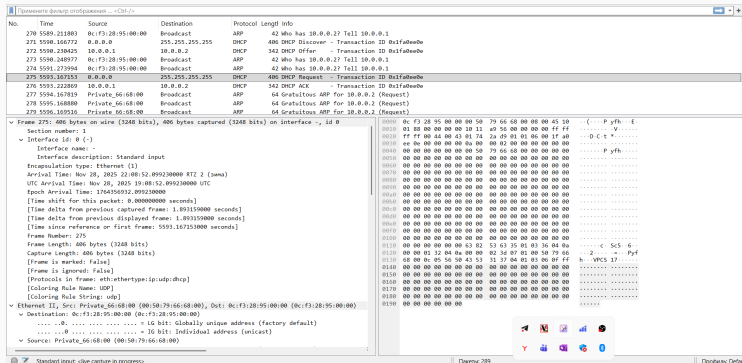


Рис. 21: анализ трафика



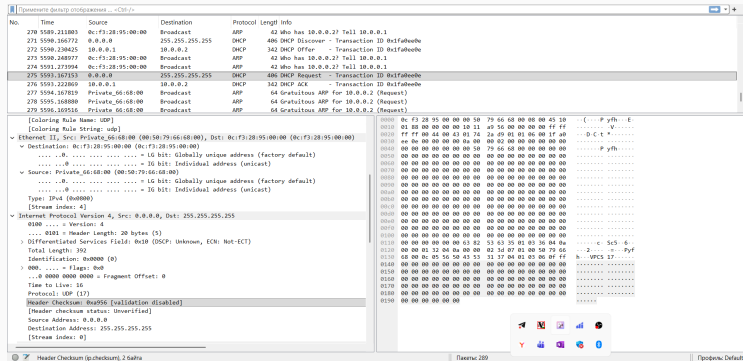


Рис. 22: анализ трафика

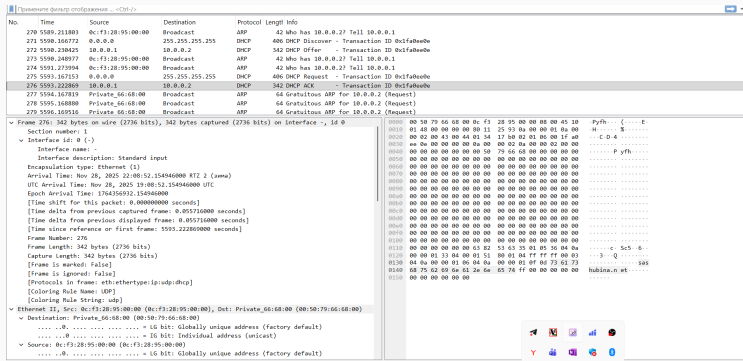


Рис. 23: анализ трафика

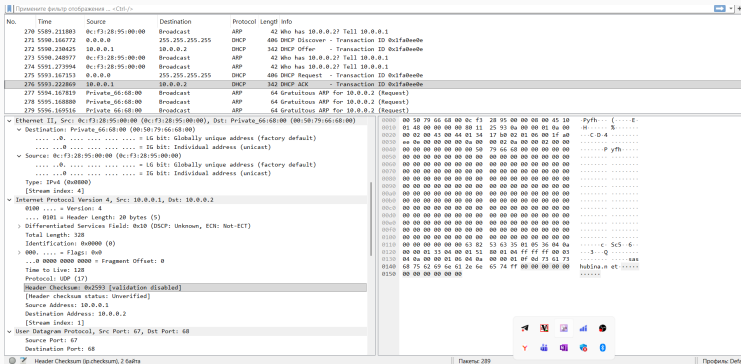


Рис. 24: анализ трафика

Примените фильтр отображения ... «Ctrl+Y»

| No. | Time        | Source            | Destination     | Protocol | Length | Info                                      |
|-----|-------------|-------------------|-----------------|----------|--------|-------------------------------------------|
| 270 | 5580.218801 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42     | Who has 10.0.0.2? Tell 10.0.0.1           |
| 271 | 5590.166772 | 0.0.0.0           | 255.255.255.255 | DHCP     | 486    | DHCP Discover - Transaction ID 0x1afade0e |
| 272 | 5590.230425 | 10.0.0.1          | 10.0.0.2        | DHCP     | 342    | DHCP Offer - Transaction ID 0x1afade0e    |
| 273 | 5590.248977 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42     | Who has 10.0.0.2? Tell 10.0.0.1           |
| 274 | 5591.273984 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42     | Who has 10.0.0.2? Tell 10.0.0.1           |
| 275 | 5593.167153 | 0.0.0.0           | 255.255.255.255 | DHCP     | 486    | DHCP Request - Transaction ID 0x1afade0e  |
| 276 | 5593.222869 | 10.0.0.1          | 10.0.0.2        | DHCP     | 342    | DHCP ACK - Transaction ID 0x1afade0e      |
| 277 | 5594.167819 | Private_66:68:00  | Broadcast       | ARP      | 64     | Gratuitous ARP for 10.0.0.2 (Request)     |
| 278 | 5595.168880 | Private_66:68:00  | Broadcast       | ARP      | 64     | Gratuitous ARP for 10.0.0.2 (Request)     |
| 279 | 5596.160516 | Private_66:68:00  | Broadcast       | ARP      | 64     | Gratuitous ARP for 10.0.0.2 (Request)     |

UDP payload (300 bytes)

Dynamic Host Configuration Protocol (ACK)

Message type: Boot Reply (2)

Hardware type: Ethernet (0x01)

Hardware address length: 6

Hops: 0

Transaction ID: 0x1afade0e

Seconds elapsed: 0

Bootp flags: 0x0000 (Unicast)

Client IP address: 10.0.0.2

Your (client) IP address: 10.0.0.2

Next server IP address: 0.0.0.0

Relay agent IP address: 0.0.0.0

Client MAC address: Private\_66:68:00 (00:50:79:66:68:00)

Client hardware address padding: 000000000000000000000000

Server host name not given

Boot file name not given

Magic cookie: DHCP

Option: (53) DHCP Message Type (ACK)

Option: (54) DHCP Server Identifier (10.0.0.1)

Option: (51) IP Address Lease Time

Option: (1) Subnet Mask (255.255.255.0)

Option: (3) Router

Option: (6) Domain Name Server

Option: (15) Domain Name

Option: (255) End

Padding: 00000000000000000000000000000000

Header Checksum (ip checksum), 2 bytes

Пакеты: 289

Профиль: Default

0000 00 50 79 66 68 00 0c f3 28 95 00 00 00 45 10 Pyfh... (.....E-  
0010 01 48 00 00 00 00 00 11 25 01 0a 00 00 01 0a 00 H.....S.....  
0020 00 02 00 43 00 44 01 34 17 b0 02 01 00 00 1f 40 ...C D 4 .....  
0030 ee 0a 00 00 00 00 0a 00 00 02 0a 00 00 02 00 00 ..... P yfh.....  
0040 00 00 00 00 00 00 50 79 66 68 00 00 00 00 00 .....  
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
00e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
00f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
0100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
0110 00 00 00 00 00 00 63 82 53 63 35 01 05 36 04 0a .....c:Sc5-6-  
0120 00 00 01 33 04 00 01 51 00 01 04 ff ff ff 00 03 ...3...Q.....  
0130 04 0a 00 00 01 06 04 0a 00 00 01 0f 04 75 61 73 .....:..sAB  
0140 68 75 62 09 6e 61 2a 6e 65 74 ff 00 00 00 00 .....hubina-n et.....  
0150 00 00 00 00 00 00

Рис. 25: анализ трафика

28/63

| No. | Time                                           | Source            | Destination     | Protocol | Length | Info                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|------------------------------------------------|-------------------|-----------------|----------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 270 | 5589.211893                                    | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42     | Who has 10.0.0.2? Tell 10.0.0.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 271 | 5590.166772                                    | 0.0.0.0           | 255.255.255.255 | DHCP     | 486    | DHCP Discover - Transaction ID 0x1fa8ee0e                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 272 | 5590.210425                                    | 10.0.0.1          | 10.0.0.2        | DHCP     | 342    | DHCP Offer - Transaction ID 0x1fa8ee0e                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 273 | 5590.248977                                    | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42     | Who has 10.0.0.2? Tell 10.0.0.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 274 | 5591.273994                                    | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42     | Who has 10.0.0.2? Tell 10.0.0.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 275 | 5593.167153                                    | 0.0.0.0           | 255.255.255.255 | DHCP     | 486    | DHCP Request - Transaction ID 0x1fa8ee0e                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 276 | 5593.222869                                    | 10.0.0.1          | 10.0.0.2        | DHCP     | 342    | DHCP ACK - Transaction ID 0x1fa8ee0e                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 277 | 5594.167819                                    | Private_66:68:00  | Broadcast       | ARP      | 64     | Gratuitous ARP for 10.0.0.2 (Request)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 278 | 5595.168880                                    | Private_66:68:00  | Broadcast       | ARP      | 64     | Gratuitous ARP for 10.0.0.2 (Request)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 279 | 5596.169516                                    | Private_66:68:00  | Broadcast       | ARP      | 64     | Gratuitous ARP for 10.0.0.2 (Request)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| v   | [Stream index: 0]                              |                   |                 |          |        | 0000 ff ff ff ff ff ff 00 50 79 66 68 00 00 00 45 10 .....P yfh...E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|     | [Stream Packet Number: 263]                    |                   |                 |          |        | 0010 01 00 00 00 00 00 10 11 a9 56 00 00 00 ff ff .....V.....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|     | > [Timestamps]                                 |                   |                 |          |        | 0020 ff ff 00 44 00 43 01 74 7a 73 01 01 00 00 1f a0 ..B-C-t z.....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|     | UDP payload (364 bytes)                        |                   |                 |          |        | 0030 ee 0e 00 00 00 00 00 00 00 00 00 00 00 00 00 .....P yfh.....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|     | Dynamic Host Configuration Protocol (Discover) |                   |                 |          |        | 0040 00 00 00 00 00 00 50 79 66 68 00 00 00 00 00 .....<br>0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>00e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>00f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>0100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>0110 00 00 00 00 00 00 63 82 53 63 35 01 01 0c 05 56 .....C-Sc5...V<br>0120 50 43 53 11 3d 07 01 00 50 79 66 68 00 ff 00 00 PCS...Pyfh...<br>0130 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>0140 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>0150 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>0160 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>0170 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>0180 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....<br>0190 00 00 00 00 00 00 00 |

Рис. 27: анализ трафика

| Примените фильтр отображения ... «Ctrl-F»                |             |                   |                 |          |                                               |
|----------------------------------------------------------|-------------|-------------------|-----------------|----------|-----------------------------------------------|
| No.                                                      | Time        | Source            | Destination     | Protocol | Length Info                                   |
| 270                                                      | 5589.211893 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42 Who has 10.0.0.2? Tell 10.0.0.1            |
| 271                                                      | 5590.166772 | 0.0.0.0           | 255.255.255.255 | DHCP     | 486 DHCP Discover - Transaction ID 0a1fa8eeb6 |
| 272                                                      | 5590.230425 | 10.0.0.1          | 10.0.0.2        | DHCP     | 342 DHCP Offer - Transaction ID 0a1fa8eeb6    |
| 273                                                      | 5590.248977 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42 Who has 10.0.0.2? Tell 10.0.0.1            |
| 274                                                      | 5591.273994 | 0c:f3:28:95:00:00 | Broadcast       | ARP      | 42 Who has 10.0.0.2? Tell 10.0.0.1            |
| 275                                                      | 5593.167353 | 0.0.0.0           | 255.255.255.255 | DHCP     | 486 DHCP Request - Transaction ID 0a1fa8eeb6  |
| 276                                                      | 5593.222869 | 10.0.0.1          | 10.0.0.2        | DHCP     | 342 DHCP ACK - Transaction ID 0a1fa8eeb6      |
| 277                                                      | 5594.167819 | Private_66:68:00  | Broadcast       | ARP      | 64 Gratuitous ARP for 10.0.0.2 (Request)      |
| 278                                                      | 5595.168880 | Private_66:68:00  | Broadcast       | ARP      | 64 Gratuitous ARP for 10.0.0.2 (Request)      |
| 279                                                      | 5596.169316 | Private_66:68:00  | Broadcast       | ARP      | 64 Gratuitous ARP for 10.0.0.2 (Request)      |
| UDP payload (100 bytes)                                  |             |                   |                 |          |                                               |
| Dynamic Host Configuration Protocol (Offer)              |             |                   |                 |          |                                               |
| Message type: Boot Reply (2)                             |             |                   |                 |          |                                               |
| Hardware type: Ethernet (0x01)                           |             |                   |                 |          |                                               |
| Hardware address length: 6                               |             |                   |                 |          |                                               |
| Hops: 0                                                  |             |                   |                 |          |                                               |
| Transaction ID: 0a1fa8eeb6                               |             |                   |                 |          |                                               |
| Seconds elapsed: 0                                       |             |                   |                 |          |                                               |
| Bootp flags: 0x0000 (Unicast)                            |             |                   |                 |          |                                               |
| Client IP address: 0.0.0.0                               |             |                   |                 |          |                                               |
| Your (client) IP address: 10.0.0.2                       |             |                   |                 |          |                                               |
| Next server IP address: 0.0.0.0                          |             |                   |                 |          |                                               |
| Relay agent IP address: 0.0.0.0                          |             |                   |                 |          |                                               |
| Client MAC address: Private_66:68:00 (00:50:79:66:68:00) |             |                   |                 |          |                                               |
| Client hardware address padding: 00000000000000000000    |             |                   |                 |          |                                               |
| Server host name not given                               |             |                   |                 |          |                                               |
| Boot file name not given                                 |             |                   |                 |          |                                               |
| Magic cookie: DHCP                                       |             |                   |                 |          |                                               |
| Option: (53) DHCP Message Type (Offer)                   |             |                   |                 |          |                                               |
| Option: (56) DHCP Server Identifier (10.0.0.1)           |             |                   |                 |          |                                               |
| Option: (51) IP Address Lease Time                       |             |                   |                 |          |                                               |
| Option: (1) Subnet Mask (255.255.255.0)                  |             |                   |                 |          |                                               |
| Option: (3) Router                                       |             |                   |                 |          |                                               |
| Option: (6) Domain Name Server                           |             |                   |                 |          |                                               |
| Option: (15) Domain Name                                 |             |                   |                 |          |                                               |
| Option: (255) End                                        |             |                   |                 |          |                                               |
| Padding: 00000000000000000000000000                      |             |                   |                 |          |                                               |
| Header Checksum (ip.checksum: 2 6a8a)                    |             |                   |                 |          |                                               |
| Пакеты: 289                                              |             |                   |                 |          |                                               |
| Протокол: 289                                            |             |                   |                 |          |                                               |

Рис. 28: анализ трафика

## Настройка DHCP в случае IPv6\*\*

---



1. В предыдущем проекте, в рабочем пространстве, я дополню сеть, размещу и соединю устройства в соответствии с топологией. Я буду использовать хост (клиент) Kali Linux CLI (добавлю образ Kali Linux CLI в перечень устройств в GNS3), поскольку клиент VPCS не поддерживает DHCPv6.

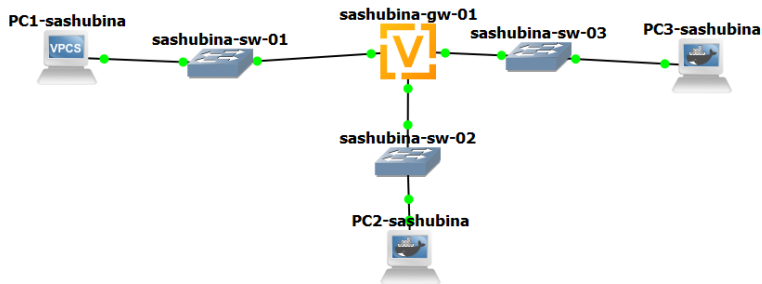


Рис. 29: создание проекта

# На маршрутизаторе я настрою адресацию IPv6:

```
sashubina@sashubina-gw-01:~$ configure
(edit)
sashubina@sashubina-gw-01# set interfaces ethernet eth1 address 2000::1/64
(edit)
sashubina@sashubina-gw-01# set interfaces ethernet eth2 address 2001::1/64
(edit)
sashubina@sashubina-gw-01# show interfaces
ethernet eth0 {
  address 10.0.0.1/24
  hw-id 0c:f3:28:95:00:00
}
ethernet eth1 {
+ address 2000::1/64
  hw-id 0c:f3:28:95:00:01
}
ethernet eth2 {
+ address 2001::1/64
  hw-id 0c:f3:28:95:00:02
}
ethernet eth3 {
  hw-id 0c:f3:28:95:00:03
}
ethernet eth4 {
  hw-id 0c:f3:28:95:00:04
}
ethernet eth5 {
  hw-id 0c:f3:28:95:00:05
}
ethernet eth6 {
  hw-id 0c:f3:28:95:00:06
}
ethernet eth7 {
  hw-id 0c:f3:28:95:00:07
}
ethernet eth8 {
  hw-id 0c:f3:28:95:00:08
}
ethernet eth9 {
  hw-id 0c:f3:28:95:00:09
}
loopback lo {
}
(edit)
```

Рис. 30: настройка адресации

```
[edit]  
sashubina@sashubina-gw-01# commit  
[edit]  
sashubina@sashubina-gw-01# save  
Saving configuration to '/config/config.boot'...  
Done  
[edit]  
sashubina@sashubina-gw-01# █
```

Рис. 31: настройка адресации

## Настройка DHCPv6 без отслеживания состояния (Stateless):

```
sashubina@sashubina-gw-01# set service router-advert interface eth1 prefix 2000:
:/64
[edit]
sashubina@sashubina-gw-01# set service router-advert interface eth1 other-config
-flag                                     ^C
tateless
[edit]ina@sashubina-gw-01# set service dhcpv6-server shared-network-name userna-stateless
sashubina@sashubina-gw-01# ^C
[edit]
na-statelessashubina-gw-01# set service dhcpv6-server shared-network-name sashubi
[edit]
na-stateless subnet 2000::0/64 service dhcpv6-server shared-network-name sashubi
[edit]
na-stateless common-options name-server 2000::lserver shared-network-name sashubi
[edit]
na-stateless common-options domain-search sashubina.netared-network-name sashubi
[edit]
sashubina@sashubina-gw-01# commit
[edit]
sashubina@sashubina-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
```

Рис. 32: добавление конфигурации

```
sashubina@sashubina-ge-01# run show configuration
interfaces {
    ethernet eth0 {
        address 10.0.0.1/24
        hw-id 0c:f3:28:95:00:00
    }
    ethernet eth1 {
        address 2000::1/64
        hw-id 0c:f3:28:95:00:01
    }
    ethernet eth2 {
        address 2001::1/64
        hw-id 0c:f3:28:95:00:02
    }
    ethernet eth3 {
        hw-id 0c:f3:28:95:00:03
    }
    ethernet eth4 {
        hw-id 0c:f3:28:95:00:04
    }
    ethernet eth5 {
        hw-id 0c:f3:28:95:00:05
    }
    ethernet eth6 {
        hw-id 0c:f3:28:95:00:06
    }
    ethernet eth7 {
        hw-id 0c:f3:28:95:00:07
    }
    ethernet eth8 {
        hw-id 0c:f3:28:95:00:08
    }
    ethernet eth9 {
        hw-id 0c:f3:28:95:00:09
    }
    loopback lo {
    }
}
service {
    dhcp-server {
        shared-network-name sashubina {
            domain-name sashubina.net
            name-server 10.0.0.1
            subnet 10.0.0.0/24 {
                default-router 10.0.0.1
                range hosts {
                    start 10.0.0.2
                    stop 10.0.0.253
                }
            }
        }
    }
}
```

Рис. 33: просмотр конфигурации

```
}
dhcpv6-server {
  shared-network-name sashubina-stateless {
    common-options {
      domain-search sashubina.net
      name-server 2000::1
    }
    subnet 2000::0/64 {
    }
  }
}
router-advert {
  interface eth1 {
    other-config-flag
    prefix 2000::/64 {
    }
  }
}
ssh {
}
}
system {
  config-management {
    commit-revisions 100
  }
  conntrack {
    modules {
      ftp
      h323
      nfs
      pptp
      sip
      sqlnet
      tftp
    }
  }
  console {
    device ttyS0 {
      speed 115200
    }
  }
  domain-name sashubina.net
  host-name sashubina-gw-01
  login {
    user sashubina {
      authentication {
        encrypted-password *****
      }
    }
  }
}
```

Рис. 34: просмотр конфигурации

```
    }  
  }  
  ntp {  
    server time1.vyos.net {  
    }  
    server time2.vyos.net {  
    }  
    server time3.vyos.net {  
    }  
  }  
  syslog {  
    global {  
      facility all {  
        level info  
      }  
      facility protocols {  
        level debug  
      }  
    }  
  }  
}  
[edit]  
sashubina@sashubina-gw-01#  
[edit]
```

Рис. 35: просмотр конфигурации

Перед тем как приступить к PC2, я изменила Kali Linux, необходимо расширить его, я открыла preferences, далее Docker container templates, ввела необходимые параметры, и установила эти компьютеры на место прошлых, это необходимо для корректной работы команд в консоли.

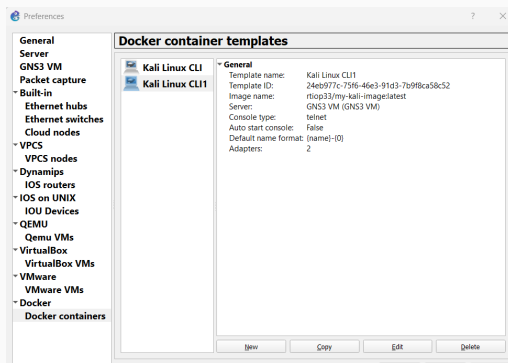


Рис. 36: установка



# На узле PC2 я проверю настройки сети:

```
[root@PC2-sashubina] ~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.42:71ff:fe9e:9200 prefixlen 44 scopeid 0x20<link>
    inet6 2000::42:71ff:fe9e:9200 prefixlen 44 scopeid 0x0<global>
    ether 02:42:71:9e:92:00 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 2000::42:71ff:fe9e:9201 prefixlen 44 scopeid 0x20<link>
    ether 02:42:71:9e:92:01 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (local loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[root@PC2-sashubina] ~# route -n -A inet6
Kernel IPv6 routing table

```

| Destination                 | Next Hop              | Flag | Met  | Ref | Use | If   |
|-----------------------------|-----------------------|------|------|-----|-----|------|
| 2000::/64                   | ::                    | U    | 256  | 1   | 0   | eth0 |
| fe80::/64                   | ::                    | U    | 256  | 1   | 0   | eth0 |
| fe80::/64                   | ::                    | U    | 256  | 1   | 0   | eth1 |
| ::/0                        | fe80::ef2:20ff:fe95:1 | UG   | 1024 | 1   | 0   | ec   |
| ::1/128                     | ::                    | U    | 0    | 3   | 0   | lo   |
| 2000::42:71ff:fe9e:9200/128 | ::                    | U    | 0    | 2   | 0   | eth0 |
| 2000::42:71ff:fe9e:9200/128 | ::                    | U    | 0    | 3   | 0   | eth0 |
| 2000::42:71ff:fe9e:9201/128 | ::                    | U    | 0    | 3   | 0   | eth1 |
| fe80::/8                    | ::                    | U    | 256  | 3   | 0   | eth0 |
| fe80::/8                    | ::                    | U    | 256  | 1   | 0   | eth1 |
| ::/0                        | ::                    | U    | -1   | 1   | 0   | lo   |

```
[root@PC2-sashubina] ~#
```

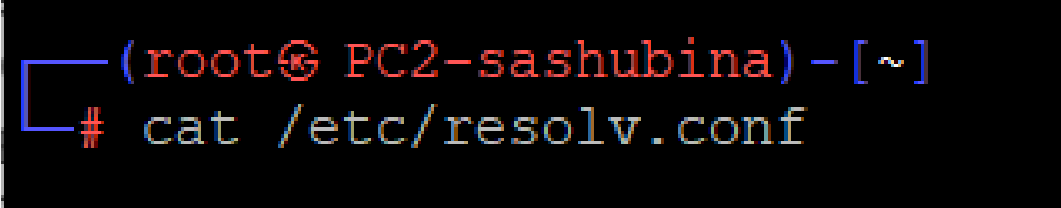
Рис. 37: проверка настроек сети

## На узле PC2 я пропингую маршрутизатор:

```
(root@PC2-sashubina)-[~]  
# ping 2000::1 -c 2  
PING 2000::1 (2000::1) 56 data bytes  
64 bytes from 2000::1: icmp_seq=1 ttl=64 time=18.6 ms  
64 bytes from 2000::1: icmp_seq=2 ttl=64 time=1.38 ms  
  
--- 2000::1 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1002ms  
rtt min/avg/max/mdev = 1.381/10.000/18.619/8.619 ms
```

Рис. 38: пинг

На узле PC2 я проверю настройки DNS:



```
(root@ PC2-sashubina) - [~]  
# cat /etc/resolv.conf
```

A terminal window with a black background. The prompt is "(root@ PC2-sashubina) - [~]" in red and blue. The command "# cat /etc/resolv.conf" is entered in blue and white.

Рис. 39: проверка настроек

## На узле PC2 я получу информацию по DHCPv6 (без запроса адреса):

```
(root@ PC2-sashubina)-[~]  
# dhclient -6 -S -v eth0  
Internet Systems Consortium DHCP Client 4.4.3-P1  
Copyright 2004-2022 Internet Systems Consortium.  
All rights reserved.  
For info, please visit https://www.isc.org/software/dhcp/  
  
Listening on Socket/eth0  
Sending on Socket/eth0  
Created duid "\000\003\000\001\002Bq\236\222\000".  
PRC: Requesting information (INIT).  
XMT: Forming Info-Request, 0 ms elapsed.  
XMT: Info-Request on eth0, interval 930ms.  
RCV: Reply message on eth0 from fe80::ef3:28ff:fe95:1.  
PRC: Done.
```

Рис. 40: получим информацию

# Я вновь пропингую от узла PC2 маршрутизатор и проверю настройки DNS:

```
(root@ PC2-sashubina)-[~]
# ping 2000::1 -c2
PING 2000::1 (2000::1) 56 data bytes
64 bytes from 2000::1: icmp_seq=1 ttl=64 time=2.27 ms
64 bytes from 2000::1: icmp_seq=2 ttl=64 time=1.23 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.234/1.753/2.272/0.519 ms

(root@ PC2-sashubina)-[~]
# cat /etc/resolv.conf
search sashubina.net.
nameserver 2000::1

(root@ PC2-sashubina)-[~]
#
```

Рис. 41: пинг и проверка настроек

На маршрутизаторе я посмотрю статистику DHCP-сервера и выданные адреса:

```
sashubina@sashubina-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication      Lease expiration      Remaining
Type      Pool      IAID_DUID
-----
[edit]
```

Рис. 42: просмотр статистики

также проанализирую захваченные анализатором трафика пакеты, относящиеся к работе DHCPv6.

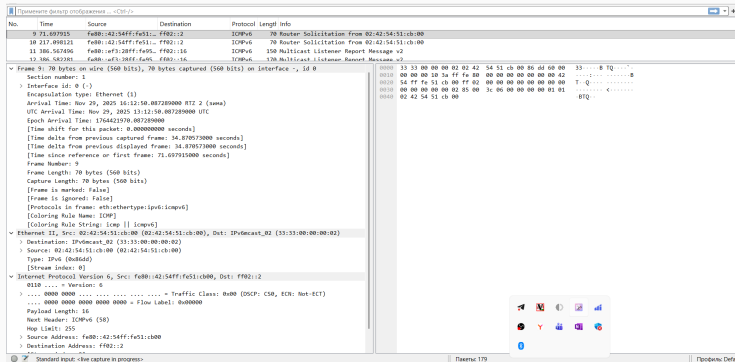


Рис. 43: анализ трафика

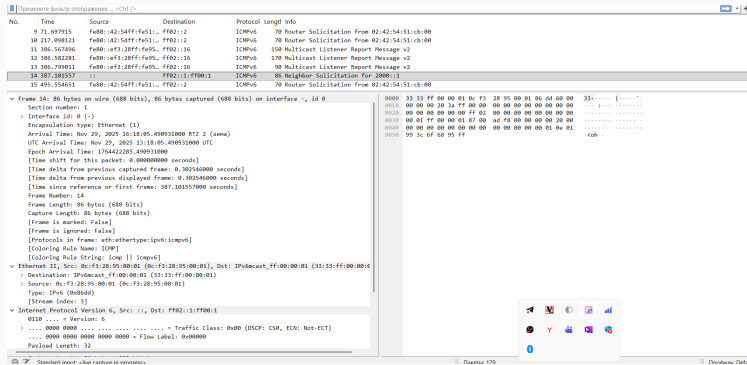


Рис. 44: анализ трафика



# Настройка DHCPv6 с отслеживанием состояния (Stateful):

```
sashubina@sashubina-gw-01# set service router-advert interface eth2 managed-flag
Cedit]
sashubina@sashubina-gw-01# set service dhcpv6-server shared-network-name userna-stateful
[edit]ful
na-statefulsashubina-gw-01# set service dhcpv6-server shared-network-name sashubi
[edit]
sashubina@sashubina-gw-01# set service dhcpv6-server shared-network-name sashubi
na-stateful subnet 2001::0/64
[edit]
sashubina@sashubina-gw-01# set service dhcpv6-server shared-network-name sashubi
na-stateful subnet 2001::0/64 name-server 2001::1
[edit]
sashubina@sashubina-gw-01# set service dhcpv6-server shared-network-name sashubi
na-stateful subnet 2001::0/64 domain-search sashubina.net
[edit]
sashubina@sashubina-gw-01# set service dhcpv6-server shared-network-name sashubi
na-stateful subnet 2001::0/64 address-range start 2001::100 stop 2001::199
[edit]
sashubina@sashubina-gw-01# commit
[edit]
sashubina@sashubina-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
sashubina@sashubina-gw-01# █
```

Рис. 45: добавление конфигурации

На маршрутизаторе я посмотрю статистику DHCP-сервера и выданные адреса:

```
sashubina@sashubina-gw-01# run show dhcpv6 server leases
IPv6 address  State    Last communication  Lease expiration  Remaining  Type  Pool  IAID_DUID
-----
[edit]
```

Рис. 46: просмотр статистики

# Я подключусь к узлу РС3 и проверю настройки сети:

```
l-# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::42:4aff:fe46:2400 prefixlen 64 scopeid 0x20<link>
    ether 02:42:4a:46:24:00 txqueuelen 1000 (Ethernet)
    RX packets 6 bytes 544 (544.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 15 bytes 1146 (1.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::42:4aff:fe46:2401 prefixlen 64 scopeid 0x20<link>
    ether 02:42:4a:46:24:01 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

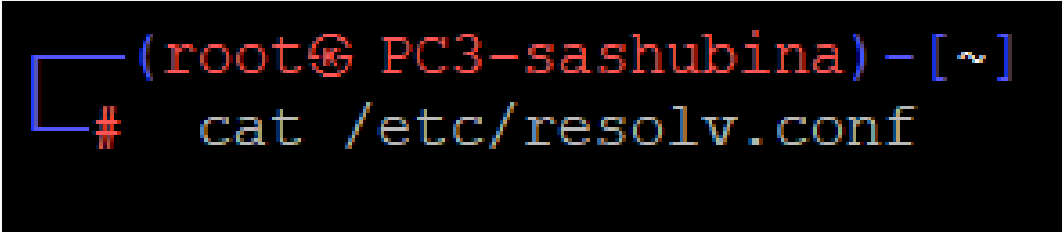
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[root@PC1-sashubina]~#
[root@PC1-sashubina]~# route -n -A inet6
Kernel IPv6 routing table
Destination      Next Hop        Flag Met Ref Use If
fe80:::/64       ::              U      256 1   0 eth0
fe80::/64        ::              U      256 1   0 eth1
::/0             fe80::e3:28ff:fe95:2 UGDAm 1024 1   0 et
ho
::1/128          ::              Un     0 2   0 lo
fe80::42:4aff:fe46:2400/128 ::              Un     0 3   0 eth0
fe80::42:4aff:fe46:2401/128 ::              Un     0 2   0 eth1
ff00::/8         ::              U      256 3   0 eth0
ff00::/8         ::              U      256 1   0 eth1
::/0             ::              In     -1 1   0 lo

[root@PC3-sashubina]~#
```

Рис. 47: просмотр настроек

На узле PC3 я проверю настройки DNS:



```
(root@PC3-sashubina) - [~]  
# cat /etc/resolv.conf
```

A terminal window with a black background. The prompt is "(root@PC3-sashubina) - [~]" in red. A blue cursor is positioned at the start of the next line, which contains the command "# cat /etc/resolv.conf" in red.

Рис. 48: просмотр настроек

# На узле РС3 я получаю адрес и конфигурацию по DHCPv6:

```
root@PC3-sashobitka:~#  
root@PC3-sashobitka:~# dhclient -6 -v eth0  
Internet Systems Consortium DHCP Client 4.4.3-P1  
Copyright 2004-2023 Internet Systems Consortium.  
All rights reserved.  
For info, please visit https://www.isc.org/software/dhcp/  
  
Listening on Socket/eth0  
Sending on Socket/eth0  
Created uid *10000010000001001270\237\002B3FF\000*.  
PBC Soliciting for leases (INIT).  
XMT: Forming Solicit, 0 ms elapsed.  
XMT: X-- IA_NA 4a:46:24:00  
XMT: | X-- Request renew in +3600  
XMT: | X-- Request rebind in +5400  
XMT: Solicit on eth0, interval 1540ms.  
RCV: Advertisement on eth0 from fe80::ef3:28ff:fe95:12.  
RCV: X-- IA_NA 4a:46:24:00  
RCV: | X-- Starts 1764621343  
RCV: | X-- t1 - renew +0  
RCV: | X-- t2 - rebind +0  
RCV: | X-- [Options]  
RCV: | | X-- IAADDR 2001::199  
RCV: | | | X-- Preferred lifetime 27000.  
RCV: | | | X-- Max lifetime 43200.  
RCV: X-- Server ID: 00:01:00:01:30:bd:f4:b7:bc:f3:28:95:00:01  
RCV: Advertisement received.  
PBC Selecting best advertised lease.  
PBC Considering best lease.  
PBC: X-- Initial candidate 00:01:00:01:30:bd:f4:b7:bc:f3:28:95:00:01 (s: 10105,  
p: 0).  
XMT: Forming Request, 0 ms elapsed.  
XMT: X-- IA_NA 4a:46:24:00  
XMT: | X-- Requested renew +3600  
XMT: | X-- Requested rebind +5400  
XMT: | | X-- IAADDR 2001::199  
XMT: | | | X-- Preferred lifetime +7200  
XMT: | | | X-- Max lifetime +7500  
XMT: V IA_NA appended.  
XMT: Request on eth0, interval 950ms.  
RCV: Reply message on eth0 from fe80::ef3:28ff:fe95:12.  
RCV: X-- IA_NA 4a:46:24:00  
RCV: | X-- Starts 1764621344  
RCV: | X-- t1 - renew +0  
RCV: | X-- t2 - rebind +0  
RCV: | X-- [Options]  
RCV: | | X-- IAADDR 2001::199  
RCV: | | | X-- Preferred lifetime 7200.
```

Рис. 49: получение адреса

Я вновь на узле PC3 проверю настройки сети, пропингую маршрутизатор и проверю настройки DNS:

```
(root@PC3-sashubina)-[~]
# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::42:4aff:fe46:2400 prefixlen 64 scopeid 0x20<link>
    inet6 2001::199 prefixlen 128 scopeid 0x0<global>
    ether 02:42:4a:46:24:00 txqueuelen 1000 (Ethernet)
    RX packets 10 bytes 1098 (1.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 22 bytes 1898 (1.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::42:4aff:fe46:2401 prefixlen 64 scopeid 0x20<link>
    ether 02:42:4a:46:24:01 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Рис. 50: просмотр настроек

```

(root@ PC3-sashubina) - [~]
# route -n -A inet6
Kernel IPv6 routing table

```

| Destination                 | Next Hop              | Flag  | Met  | Ref | Use | If   |
|-----------------------------|-----------------------|-------|------|-----|-----|------|
| 2001::199/128               | ::                    | Ue    | 256  | 1   | 0   | eth0 |
| fe80::/64                   | ::                    | U     | 256  | 1   | 0   | eth0 |
| fe80::/64                   | ::                    | U     | 256  | 1   | 0   | eth1 |
| ::/0                        | fe80::ef3:28ff:fe95:2 | UGDAe | 1024 | 1   | 0   | eth0 |
| ::1/128                     | ::                    | Un    | 0    | 3   | 0   | lo   |
| 2001::199/128               | ::                    | Un    | 0    | 2   | 0   | eth0 |
| fe80::42:4aff:fe46:2400/128 | ::                    | Un    | 0    | 4   | 0   | eth0 |
| fe80::42:4aff:fe46:2401/128 | ::                    | Un    | 0    | 2   | 0   | eth1 |
| ff00::/8                    | ::                    | U     | 256  | 3   | 0   | eth0 |
| ff00::/8                    | ::                    | U     | 256  | 1   | 0   | eth1 |
| ::/0                        | ::                    | ln    | -1   | 1   | 0   | lo   |

```

(root@ PC3-sashubina) - [~]
# ping 2001::1 -c 2
PING 2001::1 (2001::1) 56 data bytes
64 bytes from 2001::1: icmp_seq=1 ttl=64 time=48.0 ms
64 bytes from 2001::1: icmp_seq=2 ttl=64 time=3.85 ms

--- 2001::1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 3.849/25.912/47.976/22.063 ms

(root@ PC3-sashubina) - [~]
# cat /etc/resolv.conf
search sashubina.net.
nameserver 2001::1

(root@ PC3-sashubina) - [~]

```

Рис. 51: просмотр настроек, пинг

## 19. На маршрутизаторе я посмотрю статистику DHCP-сервера и выданные адреса:

```
[edit]
sashubina@sashubina-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication    Lease expiration      Remaining
Type             Pool              IAID_DUID
-----
-----
-----
2001::199         active     2025/11/30 19:48:12    2025/11/30 21:53:12    1:56:46
non-temporary    sashubina-stateful 00:24:46:4a:00:01:00:01:30:c0:b8:9f:02:42:4a
:46:24:00
[edit]
sashubina@sashubina-gw-01#
```

Рис. 52: просмотр статистики



а также проанализирую захваченные анализатором трафика пакеты, относящиеся к работе DNSРv6 и назначению адреса устройству.

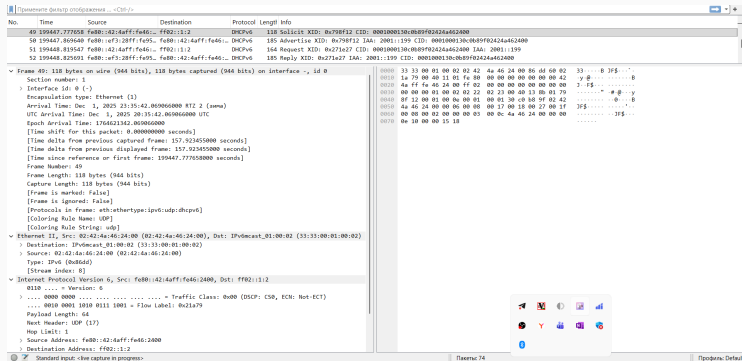


Рис. 53: анализ трафика

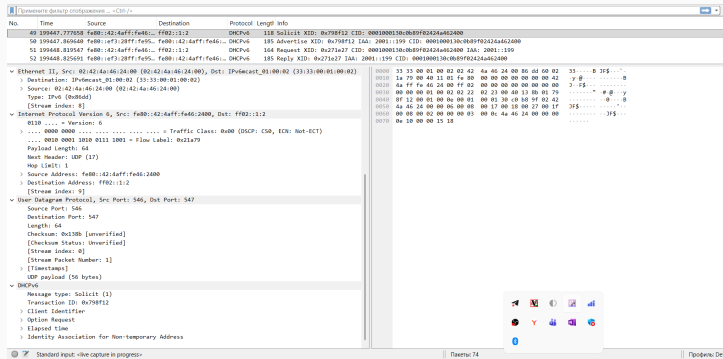


Рис. 54: анализ трафика

Примените фильтр отображения: <Ctrl>-/

| No. | Time         | Source                   | Destination             | Protocol | Length | Info                                                                     |
|-----|--------------|--------------------------|-------------------------|----------|--------|--------------------------------------------------------------------------|
| 49  | 19447.777558 | fe80::42:daff:fe46::...  | ff02::1:2               | DHCPv6   | 118    | Solicit XID: 0x798f12 CID: 0001000130:0b09f02424a462400                  |
| 50  | 19447.869540 | fe80::ef3:28ff:fe95::... | fe80::42:daff:fe46::... | DHCPv6   | 185    | Advertise XID: 0x798f12 IAA: 2001::199 CID: 0001000130:0b09f02424a462400 |
| 51  | 19448.819547 | fe80::42:daff:fe46::...  | ff02::1:2               | DHCPv6   | 164    | Request XID: 0x271e27 CID: 0001000130:0b09f02424a462400 IAA: 2001::199   |
| 52  | 19448.825091 | fe80::ef3:28ff:fe95::... | fe80::42:daff:fe46::... | DHCPv6   | 185    | Reply XID: 0x271e27 IAA: 2001::199 CID: 0001000130:0b09f02424a462400     |

▼ Frame 50: 185 bytes on wire (1480 bits), 185 bytes captured (1480 bits) on Interface -, id 0

Section number: 1

Interface id: 0 (-)

Encapsulation type: Ethernet (1)

Arrival Time: Dec 1, 2025 23:55:42.161040000 RT 2 (pna)

UTC Arrival Time: Dec 1, 2025 20:55:42.161040000 UTC

Epoch Arrival Time: 1766421342.161040000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 0.091982000 seconds]

[Time delta from previous displayed frame: 0.091982000 seconds]

[Time since reference or first frame: 199467.809600000 seconds]

Frame Number: 50

Frame Length: 185 bytes (1480 bits)

Capture Length: 185 bytes (1480 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ip:udp:dhcpv6]

[Coloring Rule Name: UDP]

[Coloring Rule String: udp]

▼ Ethernet II, Src: 0c:f3:28:95:00:02 (0c:f3:28:95:00:02), Dst: 02:42:4a:46:24:00 (02:42:4a:46:24:00)

Destination: 02:42:4a:46:24:00 (02:42:4a:46:24:00)

Source: 0c:f3:28:95:00:02 (0c:f3:28:95:00:02)

Type: IPv6 (0x86dd)

[Stream index: 9]

▼ Internet Protocol Version 6, Src: fe80::ef3:28ff:fe95:2, Dst: fe80::42:daff:fe46:2400

0110 .... = Version: 6

.... 0000 0000 .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)

.... 1011 1010 1101 0100 = Flow Label: 0xb0a0

Payload Length: 131

Next Header: UDP (17)

Hop Limit: 64

Source Address: fe80::ef3:28ff:fe95:2

Destination Address: fe80::42:daff:fe46:2400

Standard input - live capture in progress

Пакеты: 74

Пробиты: 8

Рис. 55: анализ трафика

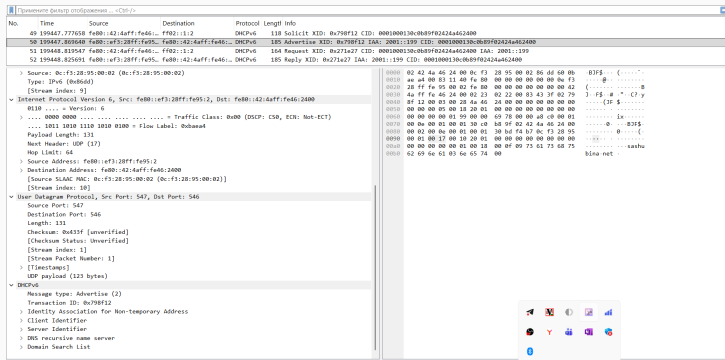


Рис. 56: анализ трафика

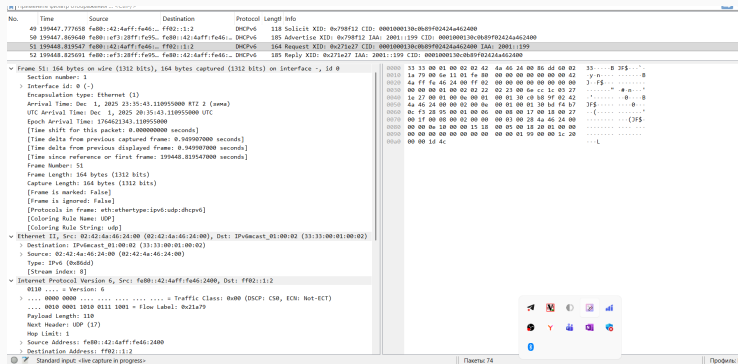


Рис. 57: анализ трафика

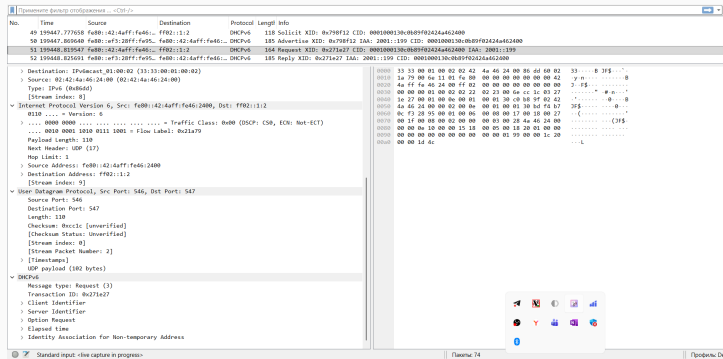


Рис. 58: анализ трафика

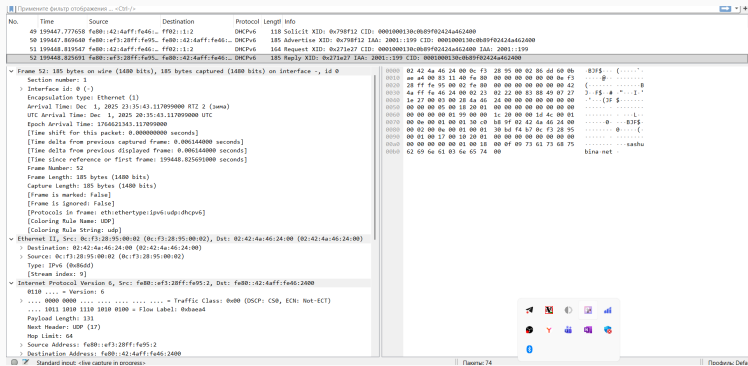


Рис. 59: анализ трафика

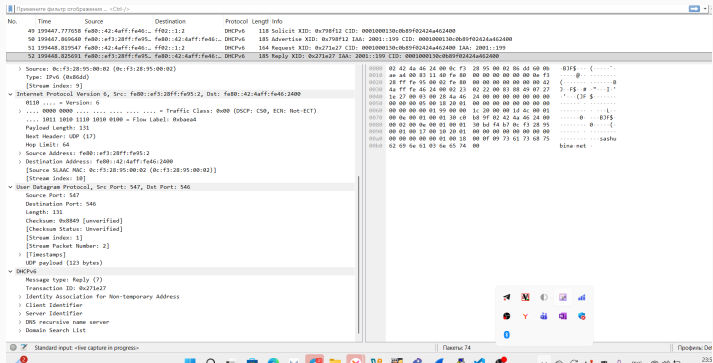


Рис. 60: анализ трафика



Я получила навыки настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6